



Implementing FPGA Framework in Now Serving Displays for Efficient Kitchen Display Systems in Delicatessens

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Abstract

The purpose of this investigation is to design a Now Serving Display circuit for a subsidiary of Dauntless Industries, LLC to track customers served by delicatessen employees before signaling employees to take a break. The scope of this research encompasses Kitchen Display System design remediation nationwide.

A Design Brief derived from the Request For Proposal and Preliminary Schematics provided will be used to virtually design and simulate a Now Serving Display circuit, on the National Instruments Multisim Circuit Design Software, prior to actual breadboarding. The final circuit prototype will be implemented on a mounted Field Programmable Gate Array framework, and every possible input will be repeatedly tested to ensure design accuracy.

The Now Serving Display circuit implemented on the Protoboard functioned with perfect accuracy and precision throughout all trials, successfully counting from 0 to 85 to signal a break for delicatessen counter employees and offering reset capabilities. This design will increase delicatessen employee satisfaction and performance, leading to decreased Kitchen Display System malfunctions and increased profits for Dauntless Industries, LLC upon commercial implementation. It is recommended that the client use the findings of this investigation to optimize Kitchen Display Systems with the designed circuit.



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1.0 Introduction

1.1 Subject

Originating in 1888, the first United States delicatessen was opened by Sussman Volk in the Lower East region of New York, named “S. Volk & Co.” However, the mid-20th century rise of fast-food restaurants, a cultural reaction to wartime regulation of resources, stifled the expansion of delicatessens beyond the Jewish diaspora. By the 1980s, public backlash to the unhealthy consumer culture created cultural promotion of healthier lifestyles focused on lean meats and salads in diet, staples of the delicatessen style, hence facilitating the proliferation of delicatessens nationwide.

Accompanying the increase in stores was an increase in customers, making the use of hand-written chits incapable of keeping up with demand. In addition, the introduction of the Fair Labor Standards Act of 1938 brought an increased focus on providing rest and meal periods to employees to bolster production efficiency across all economics sectors. Thus arose a need to develop rudimentary Kitchen Display System (KDS) circuits, both to increase the efficiency of labor through automizing customer counts, and to systematically offer breaks to employees. Further technological advancements in circuit and sensor technology have decreased the efficiency of the aforementioned circuits, resulting in the need for a more efficient replacement.

1.2 Purpose

The purpose of this investigation is to design an automated, digital Now Serving Display circuit that will keep track of the customer count served by delicatessen employees without a significant time delay to promote circuit efficiency compared to existing iterations of the circuit. The Now Serving Display circuit will stop after 85 customers to assure that employees take a break, required by a subsidiary of Dauntless Industries, LLC, managing the operation of nationwide delicatessen stores.

1.3 Scope

The scope of this research encompasses several real-life applications. With recent breakthroughs in KDSs by Independent Software Vendors (ISVs), such as Epson’s TrueOrder™ KDS, store-operating clients desire to focus on automating interactions between employees and customers. On the other hand, risks of improper record management without automation and possibilities of unnecessary safety hazards through faulty electronic systems lead to consumer dissatisfaction and hence are to be avoided. Therefore, this study’s results can be extrapolated towards Now Serving Display circuit design remediation in the United States.

However, certain limitations exist within this research investigation: only the National Instruments (NI) Digilent myDigital Protoboard (abbr. Protoboard) and materials on hand that do not consider wear-and-tear conditions (e.g. wires, additional switches, etc.) for the Now Serving Display circuit prototype are permitted, the prototype must be completed within six business days, and the prototype must be virtually simulated and approved before physical testing occurs.



1.4 Audience

This report is intended for the Directors of Panther Engineering so that the final Now Serving Display circuit device may be assessed and implemented in nationwide delicatessens under the jurisdiction of a subsidiary of Dauntless Industries, LLC. Alternatively, the findings of this study may be reviewed by the Circuits and Systems Society under the Institute of Electrical and Electronics Engineering (IEEE) to investigate greater applications of the Now Serving Display in other circuit control systems as a macro.

1.5 Development

The methods followed throughout this investigation will be presented first to provide the reader with a foundation of engineering standards followed. Any assumptions made that may impact the study will be discussed initially. The general methodology followed will be listed before the results of this study to guide the reader through the research investigation. The results will be accompanied with an analysis of the final schematic diagram created under the 'Discussion' section to justify the success of the design and will additionally allow reconstruction of this research study by the reader. Lastly, conclusions will be drawn from the Now Serving Display apparatus created and the Engineering Design Process, and references will be shown to enhance the reader's comprehension on the subject matter.

2.0 Methods, Assumptions, and Procedures

2.1 Methods

All measurements are to be taken in accordance with the *Système International d'Unités* (SI) system of measurement, regulated by the *Bureau International des Poids et Mesures* (BIPM). Metric prefixes may be used for simplicity in computations.

The NI Multisim Circuit Design Software (CDS) will be used to virtually design and simulate schematic/wiring diagrams of the Now Serving Display circuit to cement the prototype design prior to actual breadboarding. The wiring diagrams will be used to allow for seamless breadboarding of the physical prototype of the Now Serving Display circuit.

The problem was discovered through a Request for Proposal (RFP) provided by the Directors of Panther Engineering. All research conducted roughly mirrored the methodology in which civil/electrical engineering bids are solicited (through RFPs) and managed.

2.2 Assumptions

In this research investigation, functionality of the NI Multisim CDS (abbr. Multisim) in its adherence to Circuit Theory Laws and component properties is to be assumed.

Functionality of the FPGA in its ability to read and execute program files from a computer is to be assumed. This functionality is essential for the prototype to run and display the Now Serving count and hence will be assumed to facilitate progression of this research study.

Similarly, functionality of the Protoboard is to be assumed. To accurately develop the prototype for the Now Serving Display circuit, the inputs from buttons on the mounted Field Programmable Gate Array (FPGA) Programmable Logic Device (PLD) are



required in addition to the 5V Power and Ground on the Protoboard — hence, its functionality is essential to test and correct the Now Serving Display circuit. As knowledge on the Protoboard's internal mechanism workings that enable it to perform the aforementioned tasks is beyond the scope of this research investigation, it is to be assumed to allow the study to continue uninterrupted.

2.3 Procedures

1. Obtain the RFP from the Directors of Panther Engineering and read over the objectives of the project.
2. Obtain the example circuits with Multiplexed Seven Segment Displays (SSDs) from the Directors of Panther Engineering and understand circuit functionality to serve as inspiration for the Now Serving Display circuit. These example circuits will serve as the Preliminary Schematics.
3. Create a Design Brief based on the RFP to review customer requirements and make clear the criteria and constraints of the Now Serving Display circuit.
4. Modify the first example circuit to simultaneously display the two SSD outputs by increasing the clock signal frequency. This is to gain understanding of a common practice in industry that saves power while alternating the input signals at a high frequency, making the signals appear simultaneous.
5. Simulate the Now Serving Display circuit on Multisim from the example circuits while correcting the schematic design to meet all project objectives. Upon meeting all objectives, this circuit will serve as the Final Schematic.
6. Gain approval from the Directors of Panther Engineering on the Final Schematic Designs to commence the building process.
7. Create the proposed circuit on the Protoboard by mounting the FPGA PLD; proceed to carefully strip wires to meet the required length between breadboard holes, making necessary connections to the 5V Power, Ground, and SSD pins. Troubleshoot any issues in the construction process by creating workarounds via changing inputs (e.g. change from FPGA button inputs to switches) or changing circuit configuration (e.g. navigating around a breadboard row suspected to be defective).
8. Export the Multisim Final Schematic to the FPGA PLD and test the Now Serving Display circuit prototype's adherence to the project objectives. Troubleshoot any issues in meeting customer specifications by again simulating the Multisim Final Schematic to check for errors (step 4) and/or consulting the Directors of Panther Engineering in case of a hardware issue unresolved in step 6.
9. Provide a practical demonstration of the device created to the Directors of Panther Engineering to verify that all customer objectives and functionality are satisfied.
10. Draft a Technical Report presenting the findings of the study in accordance with the ANSI/NISO Z39.18-2005 (R2010) Technical Report standards.



3.0 Results and Discussion

3.1 Results

3.1.1 Overview

The Now Serving Display Circuit was prototyped per the design specifications outlined by RFP, provided by the Directors of Panther Engineering. Initially, a Design Brief was drafted from the RFP to focus the criteria and constraints of the prototype (see Appendix A). The Preliminary Schematics were modified by first implementing a Reset button to clear both SSDs and a Next button clocking the Least Significant Digit (LSD) subcircuit (see Appendix B). After the circuit inputs were working, a system using NAND implementation over standard AND-OR-Invert (AOI) logic was developed to cap the maximum count at 85 for optimization purposes. Upon gaining approval by the Directors of Panther Engineering for the Multisim Final Schematic (see Appendix C), the circuit was designed with an FPGA implemented on a conventional protoboard and tested for accuracy (see Appendices D and E).

The project was completed across the span of six business days, as described by the following subsections.

3.1.2 Day 01

The RFP from the Directors of Panther Engineering was obtained, and the objectives of the project were reviewed to gain a comprehensive understanding of the system requirements. Based on the specifications outlined in the RFP, a Design Brief with focused criteria/constraints for the prototype was developed to assure adherence to client specifications (see Appendix A). The example circuits provided by the Directors of Panther Engineering were systematically analyzed before being simulated and tested (see Appendix B). Then, the first Preliminary Schematic circuit was modified to display two numbers on the SSDs simultaneously by effectively increasing the clocking frequency along the Divide-by-Two Counters, which was mirrored in the second Preliminary Schematic.

3.1.3 Day 02

Using the second Preliminary Schematic as a base-model, a 4-Bit Synchronous Up Counter using D Flip-Flops was implemented, with its output signals being directed into the DEC_BCD_7 Integrated Circuit (IC), to control the LSD SSD. Similarly, a 74LS93 counter IC was implemented before another DEC_BCD_7 IC to control the Most Significant Digit (MSD) SSD. NAND implementation was then used to trigger the resetting of the D Flip-Flops when the count became 10, allowing for the LSD to cycle through 0-9 as intended.

3.1.4 Day 03

To ensure automatic clocking of the MSD subcircuit for every 0-9 cycle of the LSD subcircuit, the NAND implementation used to reset the LSD to 0 was inverted and then connected to the clock input of the 74LS93 counter IC. Hence, the MSD was only allowed to change when the LSD reset to 0, making the two SSDs seem as if counting in base-ten from an observer's perspective as desired. A Reset Button was then implemented that would trigger the Clear input of the D Flip-Flops and the Reset input of the 74LS93 counter IC, returning the count value to 0 in both subcircuits upon a button press.



3.1.5 Day 04

A Next Button was implemented to act as a clock signal input for the D Flip-Flops in the LSD subcircuit. AOI Logic was further added to make the D Flip-Flops reset either when the Reset Button is triggered or when the counter finishes its 0-9 cycle.

3.1.6 Day 05

To suspend the count at 85, two 4-Input AND logic gates were added, with one for each subcircuit. With the use of Inverters, each AND gate was designed to trigger only when the outputs matched the binary equivalents of 8 (1000_2) and 5 (0101_2) for the MSD and LSD subcircuits, respectively. The outputs of both AND gates were then directed to a 2-Input NAND gate. The NAND gate output was combined with the Next Button input via an AND gate, and the final output was directed to the clock signal input of the first D Flip-Flop. By doing so, the D Flip-Flop subcircuit was allowed to clock every time the Next Button was triggered until the count 85 was reached, which would set the clock signal as zero and suspend the count irrespective of subsequent Next Button presses.

3.1.7 Day 06

The Final Schematic Design was reviewed and approved by the Directors of Panther Engineering (see Appendix C). The prototype circuit was implemented on the Protoboard using a wiring diagram guideline provided by the Directors of Panther Engineering. The FPGA was placed on the conventional breadboard and its necessary inputs and outputs were wired, including an external clock connection, 5V Power, Ground, SSD output connections, and switches on the protoboard for a backup in case of push button bounces on the FPGA (see Appendices D and E). Wiring was conducted with precision, adhering to the detailed wiring diagram. Troubleshooting methods such as input adjustments and circuit reconfigurations were employed to resolve any issues encountered during the construction process. Upon successful completion, a practical demonstration as well as multiple experimental trials of the device were performed to verify functionality and compliance with the client's objectives.

3.2 Discussion

The Now Serving Display circuit implemented on the FPGA mounted onto the conventional Protoboard as described above was 100% accurate throughout all trials in meeting client specifications and functioned with precision.

While there were no issues during the simulation process, it was determined that the Protoboard used was malfunctioning on its own due to select input rows not fully functional. Furthermore, due to a lack of sufficient resources, an additional Protoboard was unable to be procured. To solve this problem, the FPGA mounted to the Protoboard was shifted down by one row on the Protoboard, and circuit connections were reestablished. With the revised setup, the circuit functioned optimally and satisfied the design specifications.

To operate the Now Serving Display circuit prototype, the Next Button, marked by BTN 0 on the FPGA, must be pressed to increase the count displayed on the SSDs by 1. Upon reaching the count of 85, the count will be suspended, rendering subsequent Next Button presses ineffective in changing the count. Then, to reset the count back to 00, the Reset Button, marked by BTN 1 on the FPGA, must be pressed.



The Now Serving Display circuit meets all the original design specifications. The prototype is wired to two common Cathode SSDs via the Protoboard. The circuit Final Schematic (see Appendix C) maintains a Multiplexed design. The LSD subcircuit is controlled by D Flip-Flops while the MSD subcircuit is controlled by a 74LS93 counter IC, both being asynchronous counters. The circuit operates on an FPGA mounted to a Protoboard Digital Mini System that has two inputs — the Next and Reset Buttons — and two outputs — the count displayed on two SSDs. Thus, the final design of the Now Serving Display circuit is justified by its perfect accuracy and precision, adeptly meeting the goals of the client.

4.0 Conclusion

The Final Design for the Now Serving Display Circuit surpassed all criteria while adhering to the limits set by the constraints. The configuration of the Final Design had yielded correct output values for all configurations of inputs (button presses); the design was built upon one NI Digilent myDigital Protoboard; and the constraint of time was also followed. The objectives set by the Directors of Panther Engineering were all accomplished in a timely manner: PLD programming principles, NAND implementation over AOI logic, application and usage of the FPGA, and circuit (broadly, engineering) design properties were learned and applied to design a Now Serving Display circuit satisfying all conditions of the original RFP/bid proposed by Dauntless Industries, LLC. To that end, the results of the Final Circuit Prototype fulfill the problem and design statements.

If given a chance, the design solution would be created differently by using external inputs (e.g. push buttons not directly on the FPGA) for operating the circuit to offer ease of use and greater precision of inputs by avoiding potential push button bounce errors. Instead of solely focusing on prototype functionality and efficiency, considering the consumer perspective in this manner would have optimized product functionality and user safety by avoiding any potential errors in the delicate FPGA, leading to greater delicatessen employee satisfaction. This would yield increased profits for the subsidiary of Dauntless Industries, LLC (which would provide greater royalty profits for Dauntless Industries, LLC's local branch with Panther Engineering) upon commercial implementation, all of which would have increased prototype success in the overall duration of this investigation.

This research study presented itself as an opportunity to achieve greater proficiency with the NI Multisim CDS, and consolidated knowledge on Circuit Design techniques, including Asynchronous Counter Timings and NAND implementation. Furthermore, an understanding of hardware inefficiencies and possibilities of error in standardized components (e.g. the NI Digilent myDigital Protoboard) was gained throughout the duration of this study.

Therefore, the Final Design was ultimately successful in surpassing all criteria/constraints and objectives placed by the subsidiary of Dauntless Industries, LLC. An efficient and cost-effective Now Serving Display circuit solution was created by this study, which can be extrapolated to real-world scenarios regarding safe and efficient automated Now Serving Display systems operated by delicatessen employees.



5.0 References

- Fact Sheet #22: Hours Worked Under the Fair Labor Standards Act (FLSA). (n.d.). DOL. <https://www.dol.gov/agencies/whd/fact-sheets/22-flsa-hours-worked>
- Maloberti, F., & Davies, A. C. (2016). A Short History of Circuits and Systems. IEEE CASS. https://ieee-cas.org/files/ieeecass/2022-01/a_short_history_of_circuits_and_systems-_ebook-_web.pdf
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- Wilson, B. (2024, October 2). Meet the Demand for a New Approach to Restaurant Kitchen Display Systems. DevPro Journal. <https://www.devprojournal.com/market-trends/restaurant/meet-the-demand-for-a-new-approach-to-restaurant-kitchen-display-systems/>



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Appendices

Appendix A: Design Brief

Client	Dauntless Industries, LLC
Target Consumer	Nationwide delicatessens operated by a subsidiary of Dauntless Industries, LLC
Designer(s)	V. Sundar
Problem Statement (restate in your own words)	Employees in delicatessens operated by a subsidiary of Dauntless Industries, LLC have made an official claim to the U.S. Department of Labor of being overworked since the opening of the delicatessens at the beginning of the year. Dauntless Industries, LLC have also suffered a loss in revenue by 20% compared to proceeds at the start of the year, which can be attributed to decreased employee efficiency. Hence, nationwide delicatessens must provide counter employees a required break after they have served 85 customers.
Design Statement	Design a digital circuit that displays the count from 00 to 85. The design requirement is for two control inputs and two output displays. The two inputs are Next and Reset. The Next signal comes from a push-button switch that, when pressed, advances the display by one. The Reset signal, which is also a push-button switch, will reset the display to a count of 00. When the display reaches 85, the count will cease. (The employee at the deli counter takes a break and a new employee takes over the counter after the 85 th customer is served.)
Criteria: How are the listed criteria measurable?	1. The two output displays are common cathode seven-segment displays. 2. The two displays require a multiplexed design. 3. The ones-unit display (Most Significant Digit) is controlled by an asynchronous counter designed with SSI logic gates (D Flip-Flops). 4. The tens-unit display (Least Significant Digit) is controlled by an asynchronous counter designed with a 74LS93 MSI counter IC. 5. The count displayed is suspended when it reaches 85. 6. The circuit must be contained on an FPGA, with two control inputs and two output displays 7. Circuit functionality will be demonstrated using a PLTW S7 PLD mounted on a Digital Mini System 8. Client will incorporate our FPGA design on to their PC board, which will include a power supply, external switches, and seven segment displays
Constraints How are the listed constraints measurable?	1. The Now Serving Display Final Schematic must be submitted for critical review on 4 April 2025, after 6 workdays. 2. The Now Serving Display circuit prototype must be demonstrated to the Directors of Panther Engineering by 4 April 2025. 3. A Technical Report must be submitted to the Directors of Panther Engineering by 4:30 PM of 4 April 2025.



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Appendix B: Preliminary Schematics

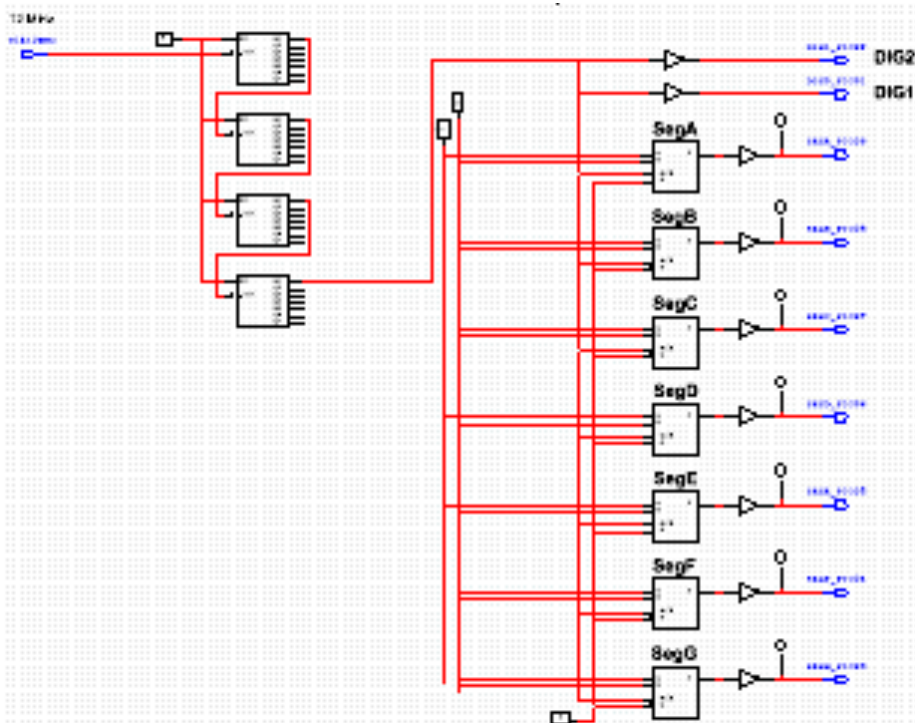


Figure 1: Simultaneously Controlled Multiplexed SSD Design Displaying Count 84

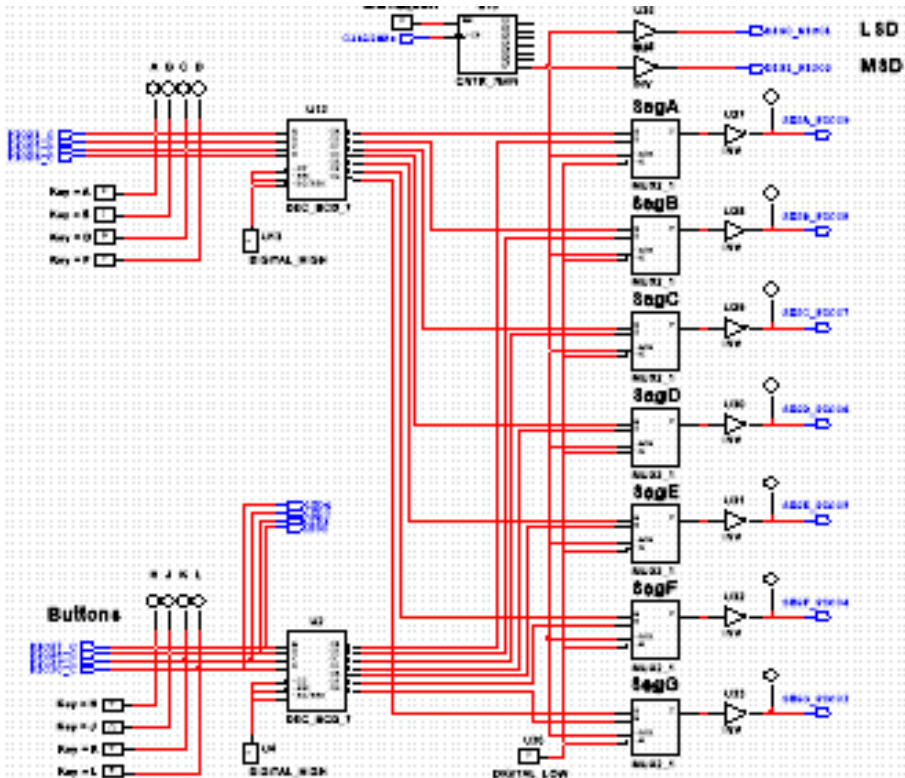


Figure 2: Independently Controlled Multiplexed SSD Design



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Appendix C: Final Schematic

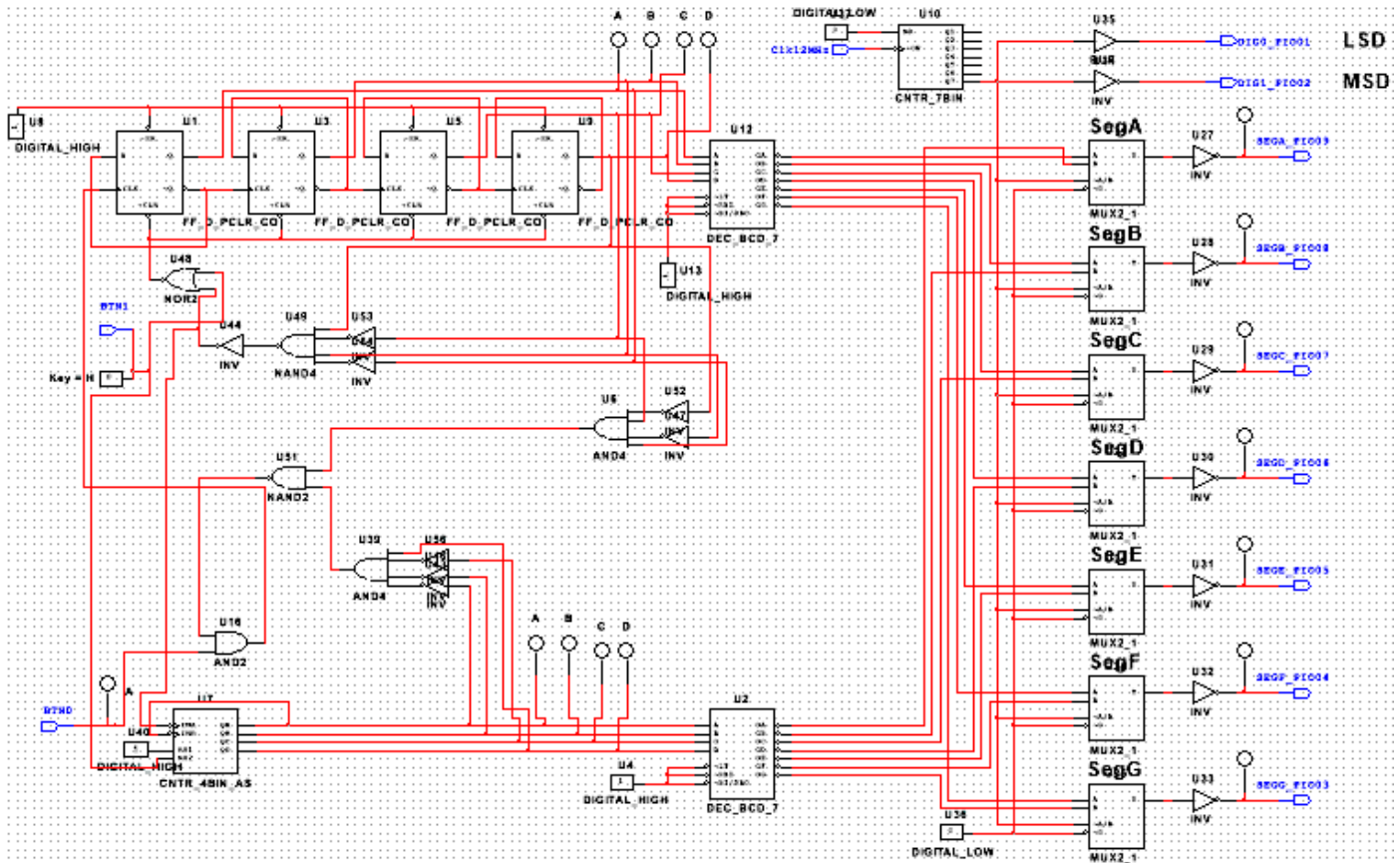


Figure 3: Now Serving Display Circuit Final Schematic



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Appendix D: Parts List

28 – Jumper Wires (Custom length)

1 – National Instruments 152190A-01 High Speed USB – 1 Meter Cable – Micro To STD Male

1 – National Instruments Digilent myDigital Protoboard

1 – National Instruments Digilent PLTW S7 Breadboardable Xilinx Spartan-7 FPGA Module



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Appendix E: Final Circuit Prototype

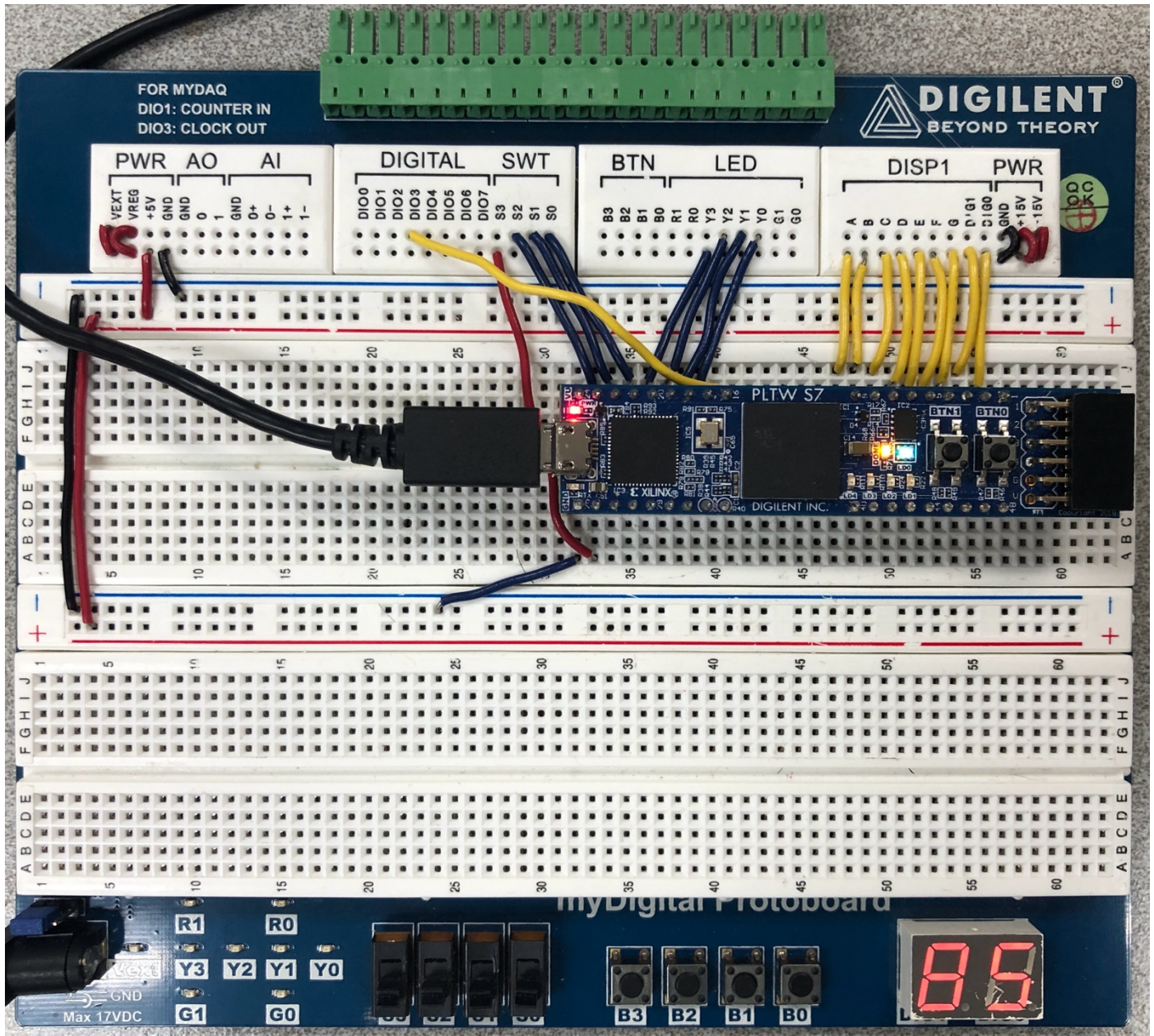


Figure 4: Now Serving Display Final Circuit Prototype



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List of Symbols, Abbreviations, and Acronyms

Symbols



Figure 5: PLD Button 0 Input Pin Schematic Symbol



Figure 6: PLD Button 1 Input Pin Schematic Symbol



Figure 7: PLD 12 MHz Internal Clock Signal Input Pin Schematic Symbol



Figure 8: PLD SSD Segment A Output Pin Schematic Symbol



Figure 9: PLD SSD Segment B Output Pin Schematic Symbol



Figure 10: PLD SSD Segment C Output Pin Schematic Symbol



Figure 11: PLD SSD Segment D Output Pin Schematic Symbol



Figure 12: PLD SSD Segment E Output Pin Schematic Symbol



Figure 13: PLD SSD Segment F Output Pin Schematic Symbol

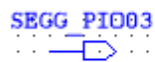


Figure 14: PLD SSD Segment G Output Pin Schematic Symbol

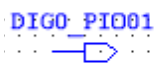


Figure 15: PLD Digital Output Pin 0 Schematic Symbol

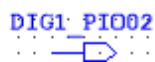


Figure 16: PLD Digital Output Pin 1 Schematic Symbol



Figure 17: PLD Digital High Voltage Common Collector (VCC) Schematic Symbol

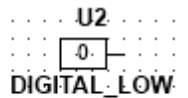


Figure 18: PLD Digital Low Ground (GND) Pin-Out Schematic Symbol

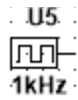


Figure 19: PLD External Digital Clock Schematic Symbol

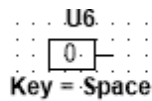


Figure 20: PLD Interactive Digital Constant Button Schematic Symbol

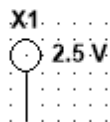


Figure 21: PLD Digital Probe Schematic Symbol

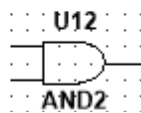


Figure 22: PLD 2-Input AND Gate Schematic Symbol

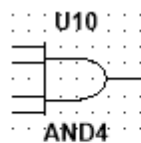


Figure 23: PLD 4-Input AND Gate Schematic Symbol

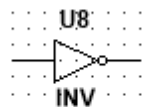


Figure 24: PLD Inverter (INV) Gate Schematic Symbol

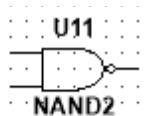


Figure 25: PLD 2-Input NAND Gate Schematic Symbol

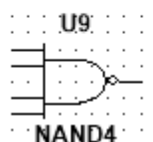


Figure 26: PLD 4-Input NAND Gate Schematic Symbol

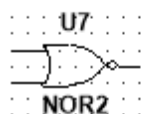


Figure 27: PLD 2-Input NOR Gate Schematic Symbol

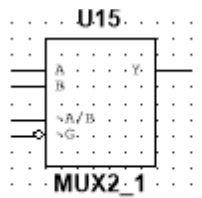


Figure 28: PLD Divide-by-Two Multiplexer Schematic Symbol

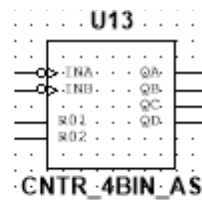


Figure 29: PLD 74LS93 MSI Counter IC Schematic Symbol

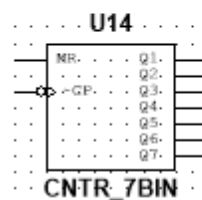


Figure 30: PLD SSD Driver Counter Schematic Symbol

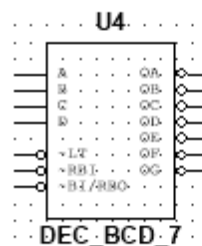


Figure 31: PLD 74LS47N Decoder Schematic Symbol

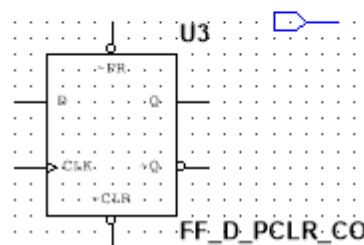


Figure 32: PLD D Flip-Flops Schematic Symbol

Abbreviations

Protoboard – National Instruments Digilent myDigital Protoboard

Multisim – National Instruments Multisim Circuit Design Software

NAND – Not-AND

NOR – Not-OR

Acronyms

ANSI – American National Standards Institute

AOI – AND-OR-Invert Logic



BIPM – International Bureau of Weights and Measures

BTN – Button

CDS – Circuit Design Software

FPGA – Field Programmable Gate Array

GND – Ground

IC – Integrated Circuit

IEEE – Institute of Electrical and Electronics Engineers

INV - Inverter

ISV – Independent Software Vendors

KDS – Kitchen Display System

LLC – Limited Liability Corporation

LSD – Least Significant Digit

MSI – Medium Scale Integration

NI – National Instruments

NISO – National Information Standards Organization

PC – Printed Circuit

PLD – Programmable Logic Device

PLTW – Project Lead The Way, Incorporated

RFP – Request For Proposal

SI – International System of Units

SSD – Seven Segment Display

SSI – Small Scale Integration

STD - Standard

USB – Universal Series Bus

VCC – Voltage Common Collector